

EDS 223: Homework Assignment 1

Relationship Between POC and Toxic Releases to Air in Warren County, NC

Karolina Sienko

2026-10-05

```
library(tidyverse)
library(sf)
library(here)
library(tmap) # for static and interactive maps

# read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data",
  "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))

# filter to a state you are interested in
nc <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "NC")

# filter to a county you are interested in
warren <- nc %>%
  dplyr::filter(CNTY_NAME %in% c("Warren County"))

# find the average values for all variables within counties
nc_counties <- aggregate(nc, by = list(nc$CNTY_NAME), FUN = mean)
```

Mapping Percentage of POC in Warren County, NC

```
tmap_mode("plot")

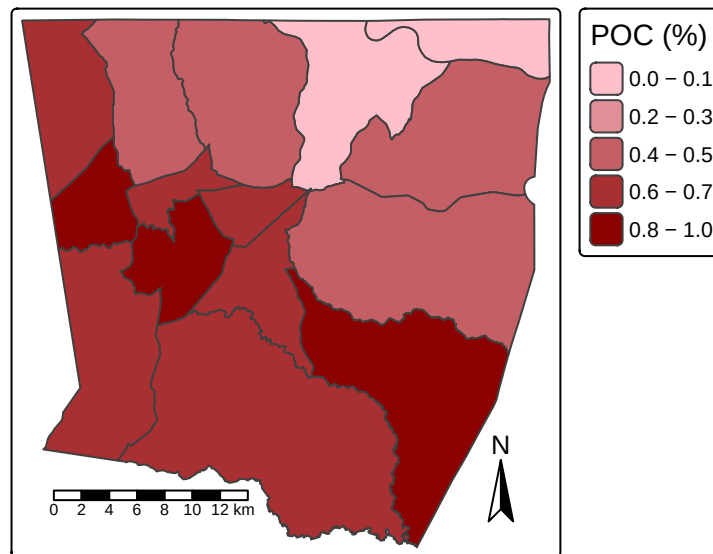
tm_shape(warren) +
  tm_polygons(size = 0.5,
    fill="PEOPCOLORPCT",
```

```

fill.scale = tm_scale(values = c("pink", "darkred")),
fill.legend = tm_legend(title = "POC (%)")
) +
tm_title("Percentage POC in Warren County, NC") +
tm_scalebar(position = c("bottom", "left")) + # positioning scalebar
tm_compass(position = c("bottom", "right"))

```

Percentage POC in Warren County, NC



Mapping Toxic Releases to Air in Warren County, NC

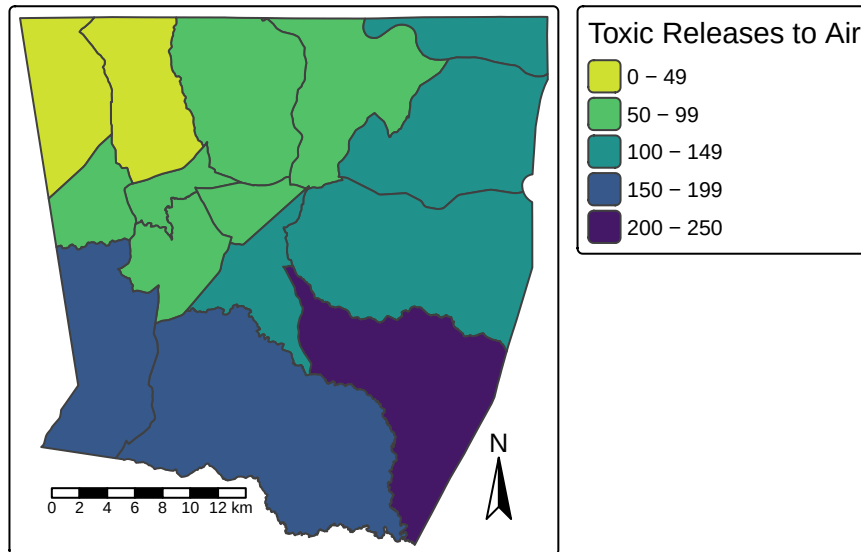
```

tmap_mode("plot")

tm_shape(warren) +
  tm_polygons(size = 0.5,
    fill="RSEI_AIR",
    fill.scale = tm_scale(values = c("-viridis")),
    fill.legend = tm_legend(title = "Toxic Releases to Air")
  ) +
  tm_title("Toxic Releases to Air in Warren County, NC") +
  tm_scalebar(position = c("bottom", "left")) + # positioning scalebar
  tm_compass(position = c("bottom", "right"))

```

Toxic Releases to Air in Warren County, NC



Analysis:

Data Citation: